Matrices and Linear Algebra notation

Lesson 4 - Assignment

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Write the following three expressions in LaTeX math display mode. All the expressions are real data science formulas.

1. A general matrix equation.

$$A\mathbf{x} = \mathbf{b}$$

where, A is a $3 \times 3 matrix, x and b are column vector of size 3.$

2. Determinant of a 2 x 2 matrix

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

3. Covariance matrix

$$\Sigma = \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i - \boldsymbol{\mu}) (\mathbf{x}_i - \boldsymbol{\mu})^T$$